Birla Institute of Technology and Science, Pilani

Instruction Division

First Semester 2026-2027

Course Handout Part II

Date: ______________________________

In addition to part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No.	CS G526
Course Title	Advanced Algorithms & Complexity
Instructor-in-Charge	Tulasimohan Molli
TA	Hans Krupakar

1. Scope and Objective

This course explores advanced paradigms in algorithm design and analysis beyond the core curriculum. Students will engage with randomized and approximation algorithms, game-theoretic techniques, and the design of parallel and distributed algorithms, with applications to modern domains such as the Internet/Web and computational biology. The course emphasizes both theoretical foundations and practical algorithm engineering.

2. Textbooks

T1 Motwani, Rajiv & P. Raghavan. *Randomized Algorithms.* CUP, 1995.
T2 Papadimitriou, C.H. & Kenneth Steiglitz. *Combinatorial Optimization: Algorithms and Complexity.* PHI, 1982.

3. Reference Books

R1 Williamson & Shmoys. *The Design of Approximation Algorithms.* Cambridge Press.
R2 Hromkovic. *Design and Analysis of Randomized Algorithms.* Springer.
R3 Vazirani, Vijay. *Approximation Algorithms.* Springer.
R4 Ausiello, G. et al. *Complexity and Approximation.* Springer.
R5 Kleinberg & Tardos. *Algorithm Design.* Pearson Education.
AR Additional reading assigned by the Instructor.

4. Course Plan

Lecs	Learning Objectives	Topics	Ref
1	Course scope and landscape	Introduction — Advanced Algorithms & Complexity	—
2	Refresher on probability and complexity classes	Review of Probability, Tail Bounds & Complexity Classes	T1 C1
5	Randomized algorithm classifications	Las Vegas & Monte Carlo, Freivalds, Min-Cut; Polynomial Identity Testing, Schwartz-Zippel	T1 C1
4	Success rates via tail inequalities	Tail Inequalities, Chebyshev, Chernoff Bounds, Applications	T1 C3-4
3	Randomized data structures	Skip Lists, Hash Tables, Cuckoo Hashing	T1 C8
2	Randomized vs deterministic graph algos	Randomized Graph: MST, Shortest Paths, Matching	T1 C10
2	Game-theoretic techniques	Game Theory: Minimax, Zero-Sum, Equilibrium, MW	T1 C4
2	Online decision-making under uncertainty	Competitive Ratio, Ski Rental, Paging, Secretary Problem	T1 C13
2	Number-theoretic algorithms	Euclid, Euler's phi, Primality Testing	T1 C14
2	Parallel and distributed algorithms	PRAM, Prefix, MIS; Byzantine Agreement, Consensus	T1 C12
2	Tutte matrix and counting	Tutte Matrix, Perfect Matchings, Isolation Lemma, #P & Counting	T1 C1, AR
6	Approximation algorithms	Greedy, LP Rounding, Primal-Dual, Local Search, Set Cover, TSP	R1, R3
4	Limits of approximation	Gap Technique, APX, PTAS, Inapproximability	R4
2	Algorithms for Web/Internet	PageRank, Web Graph, Recommendation Systems	AR
1	Algorithms for computational biology	Sequence Alignment, String Algorithms	AR
2	Student project proposal presentations	Project Proposal Presentations	—

5. Timetable

Lectures:

Day	Slot	Time	Room
Mon	7	2:00–3:00 PM	I114
Tue	5	12:00–1:00 PM	I114
Thu	5	12:00–1:00 PM	I114

Labs:

Day	Slots	Time	Room
Wed	6–7	1:00–3:00 PM	I015
Fri	6–7	1:00–3:00 PM	I015

6. Evaluation Scheme

S.No	Component	Duration	Weight	Date
1	Midterm	90 min	25%	06/10
2	Lab Interaction	During sem	10%	In class
3	Class Interaction	During sem	10%	In class
4	Term Project (Proposal + Preso + Report)	Throughout sem	20%	TBA
5	Comprehensive Exam	3 hrs	35%	04/12 AN

7. Chamber Consultation Hour

Monday 3:00–4:00 PM at H-134.

8. Make-up Policy

Prior permission of the Instructor-in-Charge is usually required to take a make-up for a test. The regulations set by AGSRD office for make-ups must be followed. A make-up test shall be granted only in genuine cases on justifiable grounds.

9. NC Criteria

A student should obtain 30% of the average of the top few performers in the class, or 40% of the median marks of the class, whichever is lower, to clear the course.

10. Notices

All notices regarding the course will be posted on Google Classroom.

11. Academic Honesty and Integrity Policy

Academic honesty and integrity are to be maintained by all the students throughout the semester. Any kind of unfair means in exams and assignments will be strictly dealt with.

Instructor-in-Charge\ CS G526